PLUMBING (Continued)

will reduce the performance of the A/C system and damage the A/C compressor. Moisture will boil at near room temperature when exposed to vacuum. A R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used to evacuate the refrigerant system. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment. To evacuate the refrigerant system, use the following procedure:

NOTE: When connecting the service equipment couplings to the refrigerant system service ports, be certain that the valve of each coupling is fully closed. This will reduce the amount of effort required to make the connection.

(1) Connect a suitable charging station, refrigerant recovery machine or a manifold gauge set with vacuum pump and refrigerant recovery equipment (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM SERVICE EQUIPMENT).

(2) Recover the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM RECOVERY).

(3) Open the suction and discharge valves and start the vacuum pump. The vacuum pump should run a minimum of 45 minutes prior to charge to eliminate all moisture within the refrigerant system. When the suction gauge reads -88 kPa (-26 in. Hg) vacuum or greater for 30 minutes, close all valves and turn off vacuum pump. If the refrigerant system fails to reach the specified vacuum, the refrigerant system likely has a leak that must be corrected. If the refrigerant system maintains specified vacuum for at least 30 minutes, start the vacuum pump, open the suction and discharge valves. Then allow the system to evacuate an additional 10 minutes.

(4) Close all valves. Turn off and disconnect the vacuum pump.

(5) Charge the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE).

REFRIGERANT SYSTEM CHARGE

WARNING: Refer to the applicable warnings and cautions for this system before performing the following operation (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - WARNINGS) and (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - CAUTIONS). Failure to follow the warnings and cautions could result in possible personal injury or death.

NOTE: Always refer to the A/C Underhood Specification Label for the refrigerant fill capacity of the vehicle being serviced.

After all refrigerant system leaks have been repaired and the refrigerant system has been evacuated, a refrigerant charge can be injected into the system. For the proper amount of the refrigerant charge, always refer to the A/C Underhood Specification Label located in the engine compartment. An R-134a refrigerant recovery/recycling/charging station that meets SAE Standard J2210 must be used to charge the refrigerant system with R-134a refrigerant. See the operating instructions supplied by the equipment manufacturer for proper care and use of this equipment.

CHARGING PROCEDURE

CAUTION: A small amount of refrigerant oil is removed from the A/C system each time the refrigerant system is recovered and evacuated. Before charging the A/C system, you MUST replenish any oil lost during the recovery process. Refer the equipment manufacturer instructions for more information.

(1) Evacuate the refrigerant system (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM EVACUATE).

(2) A manifold gauge set and a R-134a refrigerant recovery/recycling/charging station that meets SAE standard J2210 should still be connected to the A/C refrigerant system.

(3) Measure the proper amount of R-134a refrigerant and heat it to 52° C (125° F) with the charging station. See the operating instructions supplied by the equipment manufacturer for proper use of this equipment.

(4) Open both the suction and discharge valves, then open the charge valve to allow the heated refrigerant to flow into the system.

(5) When the transfer of refrigerant has stopped, close both the suction and discharge valves.

(6) If all of the refrigerant charge did not transfer from the dispensing device, open all of the windows in the vehicle and set the A/C-heater controls so that the A/C compressor is engaged and the blower motor is operating at its lowest speed setting. Run the engine at a steady high idle (about 1400 rpm). If the A/C compressor will not engage, test the compressor clutch control circuit and repair as required.

(7) Open the suction valve to allow the remaining refrigerant to transfer to the refrigerant system.